

Polar Knowledge Index: Short Demo Script (~4 min)

Team Anarch · SIH 2026 · PS 26063: Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal

1. Intro (0:00 – 0:15)

Namaste. We are Team Anarch, and this is the **Polar Knowledge Index**. Our idea in one line: every expedition, every page, every claim cited.

On screen: Home page of the app; title and tagline.

2. The Problem and the Idea (0:15 – 0:50)

NCPOR has more than forty years of Indian polar science: expedition reports, datasets, news and videos. But it sits in at least six separate systems that do not talk to each other. The old reports are scans that cannot be searched, and nothing turns them into lessons or outreach.

The Polar Knowledge Index does not ask NCPOR to type anything new. It **harvests what NCPOR already publishes**, reads it, links every item to its expedition, station, topic and year, and answers only from those pages, with the page as proof.

On screen: Scanned report on NCPOR's DSpace, then the same report inside our app, linked to its expedition.

3. What It Does (0:50 – 1:30)

Harvest: pulls expedition reports from NCPOR's DSpace archive through the standard OAI-PMH protocol, plus the NCPOR news feed, politely, one request a second, with originals left where they are.

Read: uses the PDF's own text layer, or OCR when there is none, and keeps the printed page number and the page image.

Link: ties every report, paper and news post to its expedition, station, topic and year. NCPOR's own catalogue gives confirmed links; anything inferred stays unconfirmed until a curator checks it.

Answer: hybrid search by words and by meaning. AI drafts answers, lessons and posts only from the retrieved pages, a fact-checker marks every sentence green, amber or red, and nothing is public until a named reviewer approves it.

On screen: Expedition page ISEA-9: papers grouped by NCPOR's sections, "Confirmed" and "Unconfirmed" labels.

4. Live Demo (1:30 – 2:50)

A student asks: *"Has India measured how the Dakshin Gangotri glacier is moving?"*

Search finds the Ninth Expedition paper on the glacier's secular movement, printed page 248, and opens the original scan beside its text.

The cited answer draws on the 9th, 11th, 26th and 30th Expedition reports: the snout has been monitored since 1983, and it receded 81 centimetres in 2011. All four sentences are supported.

Ask *"How fast do Mars rovers drive?"* and it refuses: no NCPOR source found.

Now a Class 11 lesson on glaciers. The checker flags four sentences red, including *"glaciers are large bodies of ice that move slowly"*: true, but not in the sources. The reviewer deletes it, approves under their own name, and publishes in English and Hindi. Every step lands in an audit log that cannot be edited.

Finally, the coverage view: reports for the 7th and 8th expeditions are listed but unpublished, and twelve later expeditions are known only from NCPOR news.

On screen: Search → page viewer → staff review of the answer (4 green) → Ask "Mars rovers" refusal → lesson with red sentences → delete, approve, publish, Hindi → coverage page.

5. Results (2:50 – 3:40)

We harvested 32 expedition reports, 765 papers and 837 news posts, with zero new data entry. We read 3,266 scanned pages from ten reports: nine already carried a text layer on most pages, and the one we fully OCR'd averaged 90 percent confidence. Every report and paper is linked to its expedition.

On a hand-verified test set, citation accuracy is [_ %] and refusal accuracy is [_ %].

Honest limits: ten of thirty reports are read in full so far, and our free local model is weaker than Gemini, so a human reviewer always has the last word.

On screen: Accuracy page with the two scores; failures listed below them.

6. Impact and Close (3:40 – 4:10)

Teachers get Indian polar lessons, students get answers they can trust, outreach staff get drafts in minutes, and NCPOR leadership sees what has never been told to the public.

It runs on a laptop with open-source tools and free AI tiers, and plugs onto what NCPOR already runs.

Polar Knowledge Index. Every expedition, every page, every claim cited.

Thank you.

On screen: *Home page, GitHub link.*

About 560 spoken words, around 4 minutes at a normal pace. Replace the two [] values with the scores from `python -m eval.run` after the test set is verified; do not record section 5 before that. Do the approval live, logged in as the reviewer.